

Systems and network biology

10

10.1 Introduction

Biological systems exhibit a high degree of complexity, which is achieved through multiple layers of hierarchy. At the heart of this assembly is the genome, which contains information necessary for the creation of molecules and the execution of a variety of processes. Following the completion of the human genome project, the genomic aspect of biology has advanced. To make use of the massive amounts of biological data generated by high-throughput techniques, novel methods for hypothesis-driven research are being developed. Now, computational methods for large-scale data analysis, as well as modeling and simulations of complex sequences, are being used. One of the most significant outcomes of the Human Genome Project was the acceleration of scientists' adoption of a new systems approach. Systems biology examines the properties and relationships between the components of a biological system during its process. This data can be combined, visualized as graphs, and modeled computationally. The Human Genome Project pioneered a new method of biology called discovery science. This defines the genetic components of humans and other organisms, views biology as a branch of information science, and provides high-throughput techniques for system understanding and the development of newer computational techniques. For more than a century, scientists have studied the individual components of cells and their functions. Despite its enormous success, it was discovered that a single molecule is rarely capable of performing a distinct biological function. The majority of the characteristics of biological systems have been discovered to be the result of complex interactions between various cellular components, including RNA, DNA, proteins, and other molecules. Thus, a significant challenge for biologists in the 21st century is to comprehend the structural and dynamic properties of this intricate web of interactions. The development of new high-throughput techniques, such as microarrays, enables us to monitor the status of cellular components at any time. Protein chips or yeast two-hybrid screening can reveal how and when these molecules interact (Altaf-Ul-Amin et al., 2014).

10.2 Network theory

The study of network theory is concerned with the representation of asymmetric or symmetric relationships between objects in a graph. In mathematics and computer science, network theory is a subfield of graph theory. Network theory is applied in a variety of fields, including physics, biology, computer science, and finance. This theory is applicable to social networks, the internet, and metabolic networks, among others. As a result of molecular interactions, various types of networks (e.g., transcription-regulatory, signaling networks) emerge. These networks cannot exist in isolation. These networks combine to form a “network of networks,” which determines the cell’s behavior. A significant challenge is to integrate theoretical and experimental data in order to map, comprehend, and model the characteristics of various networks that contribute to the behavior of these cells. The theory of complex networks has aided in our understanding of the mechanisms governing the formation and evolution of various social and technological networks. This study established that the architecture of cellular networks is comparable to that of complex systems such as society, the internet, and computer chips. This similarity demonstrated that the same laws govern the majority of complex networks found in nature, ranging from large non-biological systems to cellular systems (Barabási and Oltvai, 2004; O’Connor, 1992).

10.3 Graph theory

Leonard Euler, a mathematician, introduced graph theory for the first time. It was used to assist Königsberg in resolving a problem. Seven bridges connected four islands in this problem. Nobody could find a path that crossed all four islands and each bridge at the same time. As a result, people believed that no such path existed, but this could not be mathematically established. Euler found the solution to this problem by focusing exclusively on the relationship between the land masses and disregarding the distances and shapes of the paths. Euler demonstrated that such a path was not possible using topological features and graphs. Euler’s concept serves as the foundation for graph theory (Fig. 10.1).

In mathematics, graph theory is the study of graphs that are used to generate a model of the pairwise relationship between objects. These graphs are made up of vertices called nodes that are connected by edges called links.

Networks can be used to represent a variety of different types of data. These networks can contain a variety of different entities, such as genes or proteins. Edges represent the data associated with the connection between these nodes.

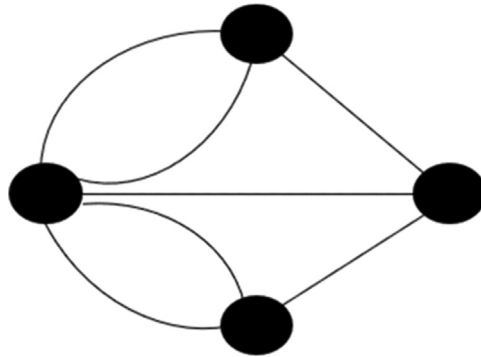


FIG. 10.1

Königsberg's problem of four islands connected by bridges.

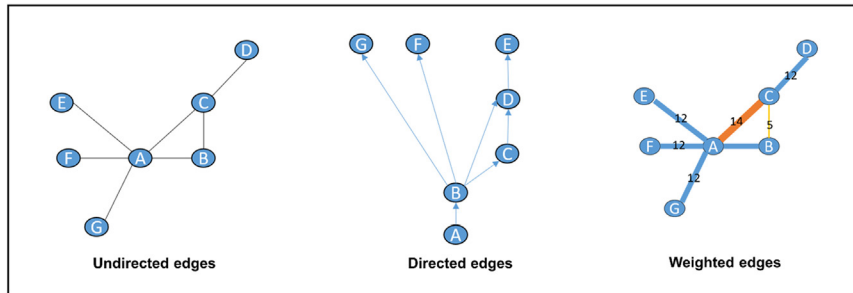


FIG. 10.2

Types of network edges.

10.4 Features of biological networks

10.4.1 The various types of network edges

Numerous types of analysis can be performed using the information provided by the edges.

There are three primary types of edges:

- (1) **Undirected Edges:** These edges represent relationships between nodes but do not indicate the direction of flow. The relationship merely demonstrates that A binds B via the evidence. These types of edges are frequently found in networks of protein–protein interactions.

- (2) **Directed Edges:** These edges are useful for hierarchically organizing a network and depict the direction of signal flow. These types of edges are frequently found in networks regulating gene expression or metabolism.
- (3) **Weighted Edges:** Weights can be applied to both undirected and directed edges. These weights can be used to display a variety of values, including sequence similarity between genes and interaction reliability. Additionally, other topological features can lend weight to the edges (Fig. 10.2)

10.4.2 Network measures

Network biology generates quantifiable networks for the purpose of characterizing various biological systems. Some fundamental network measures can be used to compare and characterize various networks.

Degree (k): A node's most fundamental property is its degree (also known as connectivity). The degree of a node indicates the number of connections it has with other nodes. For instance, Fig. 10.2 illustrates an undirected network with node A having $k = 5$. In some networks, each link has a fixed direction. In these networks, k_{in} (incoming degree) represents the number of links leading to a particular node, while k_{out} (outgoing degree) represents the number of links leading away from a node. For instance, in Fig. 10.2, Node B has a k_{out} of 4 and a k_{in} of 1. For undirected networks with L links and N nodes, the average degree is equal to $2L/N$.

Degree distribution [P(k)]: This attribute indicates the probability that a given node will have k connections. The degree distribution is calculated by dividing the total number of nodes N by the number of nodes $N(k)$. P allows for the differentiation of various classes of networks (k).

Scale-free networks: The majority of networks are scale-free, which means that their degree distribution is roughly equal to that of a power law with exponent. Hubs with lower values play a more critical role in the network. Hubs with fewer than three nodes are not considered relevant. Between 2 and 3 hubs, a hierarchy of hubs can be created. The majority of nodes in these networks are connected to a small number of neighbors. These networks' hubs contribute to their high connectivity. The hubs in Fig. 10.3A are highlighted in orange.

Path length is defined as the number of connections between two nodes. In directed networks, the smallest number of links between nodes B and A differs from the smallest number of links between nodes A and B. For instance, in Fig. 10.3B, AB equals three and BA equals one. At times, no direct path between two nodes may exist. For instance, In Fig. 10.3B, there is no path from A to C, but there is one from C to A. The average of the path lengths between all the nodes in a network is referred to as the mean path length, or $\langle l \rangle$. This metric indicates the network's overall navigability.

Coefficient of clustering or transitivity: The clustering coefficient quantifies clusters (closely linked nodes in a network). In a cluster, nodes are more connected to one another than in the rest of the network. Fig. 10.3C, illustrates the various clusters. If node X is connected to node Y and node Y is connected to node Z, there is a

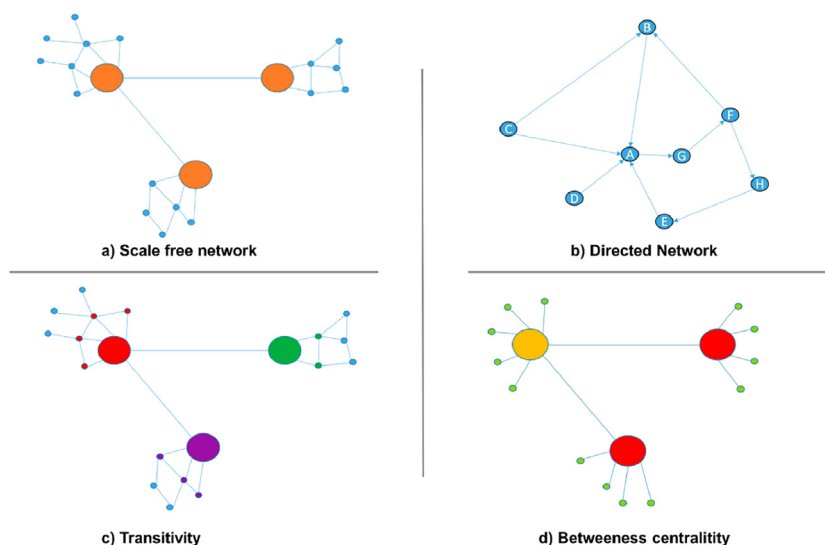


FIG. 10.3

Some network measurements.

high probability that node X is connected to node Z. This observation can be quantified using the equation $CA = 2nAk(k-1)$, where nA denotes the number of connections between kA neighbors of node A. As a result, CA indicates the total number of triangles that pass-through node A. The average clustering coefficient, or C , can be used to calculate the tendency for cluster formation within a network. $C(k)$ returns the mean clustering coefficient for each node connected by k .

Centralities: A network's centrality indicates the importance of an edge or node in terms of information flow or connectivity. Centrality is affected by a variety of factors, including a node's degree. Fig. 10.3D, illustrates the centrality of betweenness. Central nodes are colored differently, and the degree of nodes is determined by their node size (Table 10.1).

Table 10.1 The different measures of a network and their definitions.

Network measure	Definition
Degree	It indicates the number of connections from a nodes.
Degree distribution	The likelihood of a particular node having k connections.
Scale-free networks	In these networks degree distribution is approximately equal to a power law. $P(k) \sim k^{-\gamma}$
Path length	The number of links present between two nodes.
Clustering coefficient	Clustering coefficient gives a measure of clusters in a network.
Centralities	Centrality of a network predicts the importance of an edge or a node to the flow of information or to the connectivity.

10.4.3 Network models

Network models are critical for comprehending complex networks and explaining their properties. The three most critical biological networks are as follows:

Random Networks: The ER model represents a random network that begins with N nodes and assigns each pair of nodes a probability p . This generates a graph with links distributed randomly. The node degrees are then distributed according to a Poisson distribution, indicating that the majority of nodes are connected by the same number of nodes. $P(k)$ decreases incrementally in regions with high k values, indicating that the majority of nodes are present near the average value. Because $C(k)$ is not dependent on the degree of a node, it is represented by a horizontal line as k 's function. The mean path length of a network increases in logarithmic proportion to the network's size.

Scale-free Network models: These are defined by a power-law degree distribution; the probability of a node having k links can be calculated as follows:

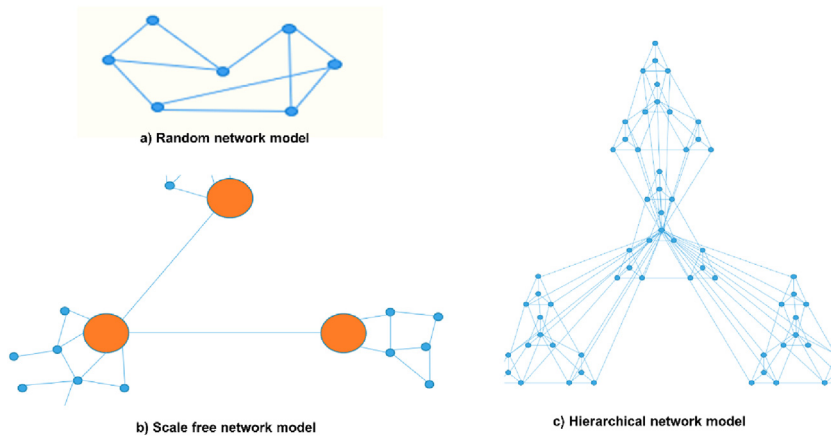
$$P(k) \sim k^{-\gamma}$$

In this case, the degree exponent equals. When compared to random graphs, highly connected nodes have a higher probability of being present in these networks. The properties of these networks are determined by hubs (small numbers of nodes with a large number of connections). According to Barabási–Albert's model of a scale-free network, the network has a power-law degree distribution defined by the degree exponent = 3. Such models lack inherent modularity; the $C(k)$ here is not dependent on k . The degree exponents of these networks, which are observed in networks, are smaller than the degree exponents of random networks.

Hierarchical network models: Hierarchical network models are iterative approaches for generating networks that may recreate the unique qualities of a scale-free topology while still exhibiting a high degree of node clustering. These qualities are abundant in nature, ranging from biology to language to certain social networks. To justify the presence of modules, it is assumed that clusters integrate in a repetitive manner, resulting in the formation of hierarchical networks. These networks are constructed using small densely connected clusters of nodes. This is followed by the generation of duplicate modules, with the external node of each cluster being connected to the central node of the previous cluster, resulting in the formation of a large node module. Identical modules are constructed, and then the peripheral nodes connected to the central node module. Nodes with low connectivity are concentrated in dense regions of the network in a hierarchical structure. These densely populated areas are linked by a few hubs (Fig. 10.4).

10.5 Types of biological networks

In a cell model, various forms are utilized to represent various types of information. The information gathered from nodes and edges varies according to the type of data utilized to construct the network. Different data types will result in different network

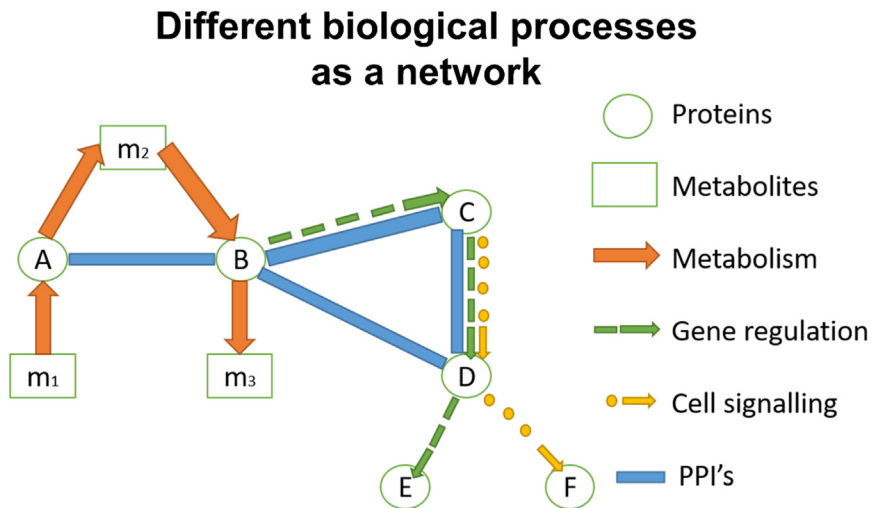
**FIG. 10.4**

Types of network models.

characteristics, including structure and connectivity. Multiple pieces of information can be conveyed in these networks via edges and nodes (Fig. 10.5).

The major classifications of biological networks are as follows:

- (1) Cell signaling networks
- (2) Gene/transcription regulation networks
- (3) Genetic interaction networks

**FIG. 10.5**

Representation of the different biological processes as a network.

- (4) Metabolic networks
- (5) Protein–protein interaction networks

10.5.1 Cell signaling networks

This network is a complex model of communication that regulates a variety of cell actions and activities. Immunity, repair, and development are all included. The cell in this network converts one type of signal to another. Serious diseases (e.g., diabetes, cancer) can be triggered by cellular signaling abnormalities. A set of metabolic events occurring within a cell and the alterations that occur as a result of receptor activation via signaling pathways. Proteins act as nodes in these networks, which are connected by directed edges.

Significant signaling pathways include the following:

MAPK pathway (mitogen-activated protein kinase): MAPK is a protein kinase enzyme that is capable of phosphorylating a variety of proteins, including transcription factors. For instance, MYC, an oncogene transcription factor, is expressed in a broad range of human malignancies. MAPK can phosphorylate MYC, hence altering the cell cycle and gene transcription. The MAPK pathway can be activated by the EGFR (Epidermal growth factor receptor). Cancer can be caused by abnormalities in the expression.

Signaling pathway of the hedgehog: This route is a critical regulator of development in animals. From flies to humans, this pathway is retained. This pathway enables the establishment of the flybody plan. During metamorphosis and development, this route is critical.

Pathway of TGF-beta signaling: This pathway is involved in cell proliferation, differentiation, and apoptosis.

10.5.2 Gene/transcription regulation networks

These networks serve as a model for the regulation of gene expression. Gene regulation is the process by which instructions contained in genes are transformed to products (such as RNA or protein). Gene regulation is responsible for the structure and function of cells. Cell differentiation, for example, can transform a cell into a specialized form. Through morphogenesis, an organism can acquire a shape.

Gene expression and repression are presented identically in these networks. Model organisms are examined in order to gain a better understanding of how other creatures function. *D. melanogaster*, *E. coli*, and *S. cerevisiae* are among these model organisms.

10.5.3 Genetic interaction networks

Genetic interactions are a cooperative phenomena in which mutations in two or more genes result in a phenotype that is different than expected when the effects of each mutation are taken together. These networks do not depict a physical connection

between the genes, but rather a functional connection. Genes are represented as nodes, and their interconnections are represented by their edges. It is possible to deduce the direction of the edges.

10.5.4 Metabolic networks

These are the biological reactions that take place within a cell. These networks are reconstructed using data from experiments and the genomic sequence. There are networks accessible for a wide variety of creatures, from bacteria to people. These networks can be used to simulate and analyze metabolism. Metabolism is a collection of biological events that assist organisms in maintaining their structure, responding to environmental changes, developing, and multiplying. Metabolic pathways are a set of chemical events occurring within the cell that are catalyzed by enzymes. This can result in the formation of a product or can initiate the beginning of further metabolic pathways. Numerous metabolic pathways coexisting in a cell might form a metabolic network. Metabolic pathways are the major responses that maintain the homeostasis of an organism. The paths between the substrates and the enzymes contain directed edges. As a result, these enzymes and substrates are represented as nodes in the network, whereas directed edges depict the reactions.

10.5.5 Protein—protein interaction networks

Protein—protein interactions are required for the majority of biological functions. As a result, understanding these PPIs is critical for understanding how cells' physiology changes during various disorders. Because drugs have an effect on these PPI networks, understanding PPIs is critical for drug development. In general, protein—protein interactions (PPIs) relate to the physical interactions or binding between proteins. These interactions are very specialized, occur in defined binding areas, and have a defined function. PPI data can be used to represent both steady and transient interactions. Protein complexes have stable interactions (e.g., hemoglobin). Modifications to proteins are temporary interactions (e.g., kinases). The interactome's dynamic component is composed of transient interactions. PPI data can be used to:

- Assign proteins functions
- Recognize minor details regarding signaling pathways
- Determine the relationships between the proteins contained in complexes such as the proteasome.

Interactome: The interactome is a term that refers to the collection of all PPIs found in a cell, an organism, or any other environment. The advancement of PPI screening tools such as the yeast two-hybrid experiment and mass spectrometry has resulted in a massive rise in the amount of PPI data and the production of more complex interactomes.

The properties of PPI networks include the following:

The small world effect in PPI networks shows that there is a high degree of interaction among the proteins. A network's diameter (the largest distance between two nodes) is always small. Between two nodes, the distance is always less than six steps. This is referred to as the "6° of separation" principle. The degree of connection is critical biologically, as it enables the rapid and efficient flow of impulses within a network. Although networks are highly coupled, alterations in a single gene or protein have little effect on the network due to the network's resiliency.

Scale-free networks: In these networks, the majority of proteins (nodes) are connected to only a few other proteins. A few proteins (hubs) are linked to a vast number of proteins (nodes) in the network. The networks can be structured in a preferential attachment approach. This model is based on the notion that when a network is built, edges are attached preferentially to nodes with the greatest degrees. This principle aids in the establishment and expansion of the network.

Due to their scale-free nature, PPI networks exhibit the following characteristics:

Stability: When a random failure occurs, the probability of harming a hub is low due to the low degree of connection of the majority of proteins. Due to the availability of multiple hubs, a network will not lose connectivity if a single hub fails.

Invariant to scale change: Regardless of the amount of nodes or edges in a network, it will always stay stable. Regardless of the network's size, hubs produce small world effects.

Vulnerability to a targeted attack: When a small number of networks in a hub fail, the network degrades into a collection of isolated graphs. The hubs contain both lethal and vital genes. For instance, some proteins associated with cancer are hub proteins (e.g., p53 protein).

Transitivity: Another critical aspect of these networks is their modularity. The clustering coefficient or transitivity of a network measures the inclination of nodes to cluster together. A higher transitivity score indicates that a collection of nodes is highly connected (communities). It is critical to locate these communities within biological networks, as they can reveal protein complexes and functional modules. The module is a term that refers to an interchangeable functional unit. These modules are network contents with known relationships to other network contents. The critical property of a module is that its internal properties remain constant regardless of the context in which it is utilized. Modules simplify biological networks by dividing them into smaller, functional components that may be perceived as a unified unit. Topological network studies can aid in locating and comprehending these modules.

A complex of proteins can be thought of as a module with stable interactions between the proteins that result in a fixed configuration at a particular time and location. These complexes are representations of multi-protein models performing prescribed functions. A broader version of these modules does not require protein binding, as their functions remain constant when used in a different setting. Additionally, knowledge of the modules aids in the comprehension of their interactions with one another and with proteins (Fig. 10.6).

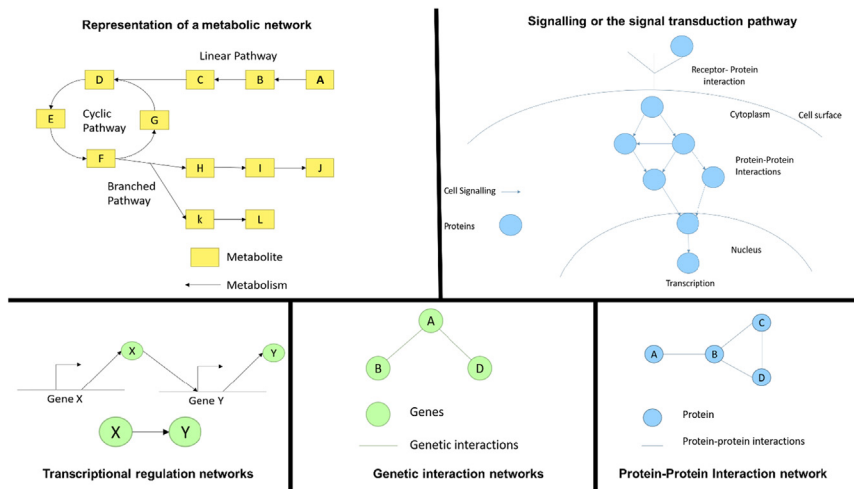


FIG. 10.6

Classifications of biological networks.

10.6 Sources of data for biological networks

The data can be obtained from different sources for generating biological networks such as:

Manual literature curation: Curators or domain experts can review previously published information and store it in databases. This can result in high-quality, well-presented data. However, curation takes a significant amount of time and money, limiting the size of databases.

High-throughput datasets: Experiments such as mass spectrometry identification and yeast two-hybrid are generating large volumes of data, such as the PPI datasets. Although large, ordered datasets are generated, the approach used introduces bias, and the quality of these datasets may vary.

Computational techniques: A variety of strategies are used to anticipate previously unknown associations between items based on current experimental evidence. For example, if orthologues exist, protein interactions in mice can be predicted by examining protein interactions in humans. This results in the creation of a tool for broadening the scope of experimental data. However, the datasets generated by these methods are noisier than the ones discussed previously.

Extracting relationships from the literature via text mining: Various machine learning techniques can be used to extract relationships from the literature. While this strategy can increase data coverage, NLP is time consuming to process and the resulting results are frequently noisy.

10.7 Gene ontology for network analysis

The initial output of genomics experiments is a list of genes that significantly alter the experimental conditions. The first step in investigating datasets is functional enrichment analysis. This determines whether the gene list is statistically enriched for particular biochemical functions and processes. The Gene Ontology (GO) consortium develops a controlled vocabulary of terms that can be used to describe genes and their products as cellular components, biological processes, and molecular functions. Utilizing the open access tools available on the GO website, an enrichment analysis could be conducted using GO. These analysis techniques examine a list of genes for the presence of GO terms that occur more frequently than expected by chance in the query list. The presence of over-represented terms indicates that unknown processes are controlled differentially and preferentially in a given condition. Both GO's strength and weakness stem from its hierarchical structure. Despite efforts to comprehend the architecture of GO enrichment analyses, determining the level of hierarchy that results in statistical enrichment is challenging. By and large, the major enriched terms encompass broader functional categories that are less useful for comprehending novel functions. Cells have pathways that are biochemical systems that aid in the conversion of signals to output. Enrichment analysis on the basis of a pathway provides information that is more pertinent and understandable in the context of critical processes occurring under a given condition. Numerous techniques for pathway analysis are available, including DAVID, InnateDB, KEGG, and Reactome; quantitative techniques rely on gene set enrichment. Current techniques attempt to take advantage of the fact that not all genes are equally capable of distinguishing between different pathways. As a result, pathway analysis can provide us with a wealth of information about previously unknown relationships between genes or pathways (Yon Rhee et al., 2008).

While pathway analysis tools are extremely powerful, they do have the following limitations:

- The majority of genes do not have a pathway annotation. For example, more than 85% of human genes in the Ensembl database are not mapped to KEGG pathways.
- There is a significant bias in favor of the more well-studied pathways.

10.8 Analysis of biological networks and interactomes

Network biology is a rapidly growing field of research that demonstrates that biochemical processes are not regulated by isolated proteins or straight pathways, but rather by a complex system of networks of interacting molecules. Understanding molecular interaction networks can aid in the development of biochemical processes and identifying the critical topological properties and nodes required for their control, which are important for explaining the complex symptoms of diseases.

According to network theory, disease phenotypes are not caused by mutations in single genes acting independently, but rather by changes in a gene's network context. Thus, understanding disease pathology and developing effective targeted therapeutics requires a thorough understanding of pathologically altered interaction networks. Network analysis is a time-efficient technique that is comparable to more traditional methods. The advantages of this technique are that it is more data intensive and less constrained by the limitations of currently used functional annotations, as a result of the existence of proteome level interactome maps for various species, such as humans. Network analyses are less biased toward well-characterized pathways and incorporate a broader variety of proteins and genes. The interactome can be visualized and comprehended by constructing a network or graph in which each entity (metabolite, miRNA, protein, or gene) is represented by a node and its interactions with other entities by edges connecting the nodes. Networks can contain a variety of nodes and edges and are frequently used to visualize and interpret a variety of different types of entities and their relationships concurrently. This can represent a highly accurate and comprehensive representation of a biological system. Attributes of nodes or edges can aid in the integration of biological data associated with the node's edges (e.g., confidence score) or (e.g., gene expression). Two distinct methods are used to conduct network analysis on a list of genes. The first method superimposes genomics data (for example, gene expression) on pre-established interaction networks (e.g., open access PPI data). The second method is to discover the network using data generated experimentally (Kovács et al., 2019).

10.9 Interaction network construction using a gene list

The type and source of data are critical considerations when constructing interaction networks. When determining the quality or type of interaction data, researchers must proceed cautiously. The IMEx (International Molecular Exchange) consortium's databases promote manual curation of experimentally generated data from the research literature. Meta-databases aggregate and repackage data from primary sources and make it accessible through a portal. A few databases incorporate computational interactions to augment the experimentally generated data. This step is critical for enriching the experimentally generated sparse interaction network. Researchers can obtain more accurate results by comparing experimentally generated data to data generated computationally. Primary interaction databases exhibit a low degree of overlap in the data they provide. This is done on purpose, as the developers of the IMEx database wish to avoid duplication in the manual curation method. This also results in the omission of supportive interaction data when a single database is used as the source. Researchers can search multiple databases concurrently using web services such as PSICQUIC (Aranda et al., 2011).

Prior to conducting network analysis, certain information must be considered. Extracted interactions can be physical (e.g., PPI), regulatory (e.g., miRNA-mRNA), or biochemical in nature (e.g., phosphorylations). While combining

different types of interactions can be beneficial, we should proceed cautiously because the edges in such a network would vary significantly, and this factor should be considered when analyzing the data. For example, PPIs typically have undirected edges and can access information about protein complexes, whereas biochemical interactions are directed and can depict data flow. Another aspect of protein interactions is that they can be detected using Mass Spectrometry and Affinity Purification, but these techniques are unable to distinguish between indirect and direct interactions, despite the fact that they are depicted as interactions in open access databases. Confidence in a specific interaction varies significantly depending on the experiment used to determine it. Massive amounts of data about the interactome could be generated using high-throughput technologies, such as yeast 2-hybrids. However, this data has a high rate of false negatives and positives. Meanwhile, interactions extracted from the literature inculcate greater confidence but favor well-characterized biological processes and pathways. Numerous methods have been developed to generate confidence scores, which are displayed on networks as edge weights. Interactomes derived from databases are a static representation of all the interactions associated with a given input list. Certain interactions are context-dependent (e.g., occurring in a particular type of cell, under certain conditions, or for a particular protein isoform). However, the literature and interaction databases are deficient in high-throughput interactome data. If this analysis is restricted to specific contexts, the vast majority of data will be omitted. Researchers are now integrating multiple types of contextual data, including gene and protein expression, to identify the most probable edges and nodes in a subnetwork.

10.10 Data analysis tools

10.10.1 The InnateDB

The InnateDB database ([Breuer et al., 2013](#)) contains a large number of experimentally verified bovine, mouse, and human interactions and pathway annotations that were compiled from publicly available interaction and pathway databases. Along with the integrated data, the InnateDB curation group has annotated a large number of immune-related reactions through a review of the biomedical literature. The interactions in InnateDB have been curated according to MIMIX standards. Additionally, InnateDB offers integrated bioinformatics tools such as ontology and pathway analysis, network analysis and visualization, as well as the ability to take and interpret user-supplied gene expression data in a pathway or network context. InnateDB also contains the entire mouse genome. In the first step, a list of genes is chosen. After that, human Ensembl gene IDs are mapped to selected gene IDs. This can be accomplished by selecting the “Data Analysis” menu and then the “Network Analysis” tab. Following that, we can upload the gene list using the “Upload Data” button. Alternatively, you can upload this data in the form of a spreadsheet or text file. More than 10 different quantities can be used to quantify data associated with genes, which can be

integrated as different node attributes in a network visualization. InnateDB provides a variety of filters for locating the interactions in the developed network. The default option returns all interactions that contain at least one gene from the gene list. This enables us to identify nodes in networks that have not been identified experimentally but are capable of interacting with these genes or proteins. The user can choose to view only interactions between the input genes or interactions within a particular pathway of interest. One can choose between predicted interactions and manually annotated interactions. The user is then presented with a table of reviews for the uploaded list. After that, the query data can be used to construct networks.

10.10.2 Visualization and download of networks

The networks generated by InnateDB can be viewed using a variety of different tools. The Innate DB results page is visualized using the CerebralWeb app (Frias et al., 2015), which layers nodes according to their subcellular location. A Cerebral plugin can be used to analyze interaction networks. Additionally, third-party software such as BioLayout Express 3D and CyOog plugins can be used. BioLayout Express 3D enables the visualization of large networks in 3D or 2D. The CyOog plugin analyses and reduces the complexity of networks by utilizing Power Graph. InnateDB networks can be downloaded in a variety of formats (Fig. 10.7).

10.10.3 Enrichr

Enrichr (Chen et al., 2013) is a web and mobile application that combines gene set libraries, a novel technique for ranking enriched terms, and multiple methods for visualizing results. Enricher is an HTML5-based application. Users are able to share

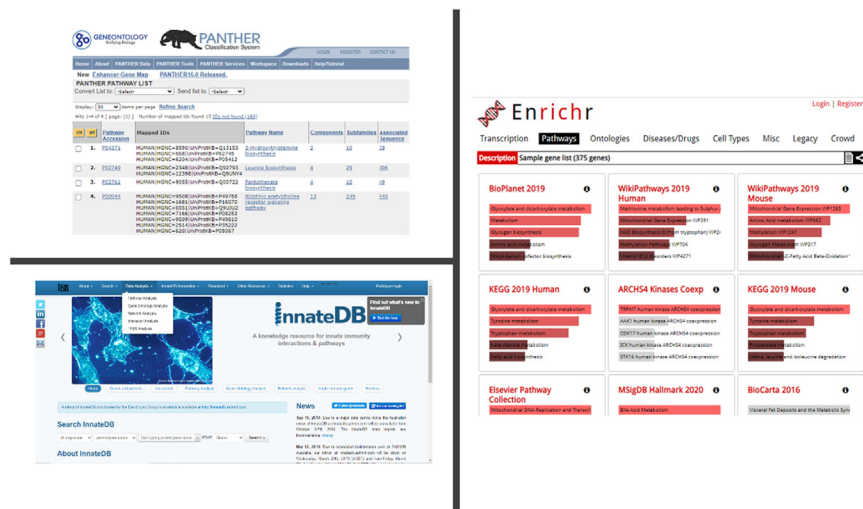


FIG. 10.7

Webpage of InnateDB, Enrichr and PANTHER.

results and view them in publication-ready formats. Enrichr is a simple web-based enrichment analysis tool that generates various visualisations for the combined functions of an input gene list (Fig. 10.7).

10.10.3.1 PANTHER

PANTHER (Thomas et al., 2022) integrates pathways, ontologies, gene functions, and statistical analysis methods to analyze large amounts of data from gene expression, proteomics, and sequencing experiments. This system utilizes 82 genomes organized into subfamilies, families, and phylogenetic trees to visualize the relationships between genes, MSA, and HMM models. Numerous classification schemes are used to classify genes, including annotated subfamilies and families, as well as sequences associated with pathways. The PANTHER website contains a variety of tools that enable users to search for and screen gene functions, as well as analyze experimental data using a variety of statistical methods (Fig. 10.7).

10.10.3.2 GSEA

Although RNA expression analysis has become a standard technique in biomedical research, extracting useful biological information remains a significant challenge. GSEA is a technique for analyzing and comprehending expression data. This technique focuses on a group of genes that share a common location, function, or regulatory role. GSEA can detect similarities in clinical data collected from patients across studies, revealing shared pathways. The GSEA technique is an open-source package that includes a database containing several standard gene sets (Fig. 10.8).

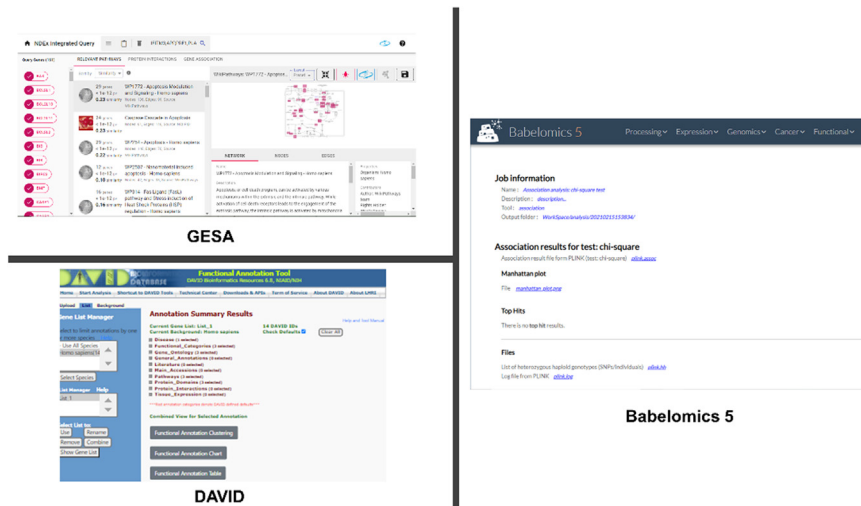


FIG. 10.8

Webpage of GESA, DAVID and Babelomics 5.

10.10.3.3 DAVID

The DAVID ([Huang et al., 2007](#)) bioinformatics resources include a knowledgebase and tools for mining biological information from large protein or gene lists. DAVID, a data mining environment, is capable of analyzing gene lists generated during genomic experiments. This process begins with the upload of a list of genes with multiple gene identifiers that are analyzed using pathway or text mining tools such as gene functional classification, functional charts, or tables. This method gives researchers with biological information contained in gene lists enriched by genome studies ([Fig. 10.8](#)).

10.10.3.4 Babelomics 5

The Babelomics 5 ([Medina et al., 2010](#)) tool accepts a list of genes or proteins and maps them to a reference interactome. The interactome can be constructed using any user-defined interactome, including human interactomes. Following the list's mapping, Babelomics 5 determines the values of several other parameters using the least connected networks and interactomes defined by proteins. Comparing separate protein lists allows for the detection of significant changes in their parameter distributions ([Fig. 10.8](#)).

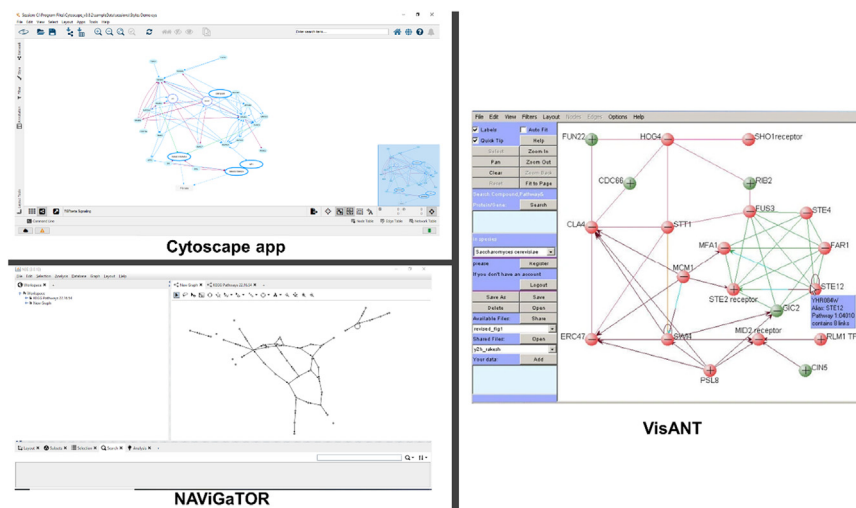
10.11 Network visualization tools

10.11.1 Cytoscape

Cytoscape ([Shannon et al., 2003](#)) is an open access environment for combining, visualizing, and querying networks. Cytoscape's main software part gives the ability to input and output data, integrates interactions, network, data visualization, filtering and querying tools. Cytoscape's VizMapper helps give visual mapping to attributes, controlling visualization of edges and nodes on the basis of their molecular states. These mappings help to overlay many data types in networks. Cytoscape was created in Java and distributed as a free software. It has been integrated with other applications (e.g., geWorkbench) and tools, websites (e.g., network image generator), databases (e.g., BIND, MiMI). Commercial companies like GeneGO, Genespring, Agilent have utilized this software. The core of Cytoscape can be expanded through plugin structures, allowing fast modeling, development of analysis and features. Many third-party programmers are involved in developing plugins for Cytoscape, due to the popularity of Cytoscape as a free environment. 74 open access plugins have been produced since 2004, 46 of these plugins are compatible with the latest versions of Cytoscape ([Fig. 10.9](#)).

10.11.2 NAViGaTOR

NAViGaTOR ([Brown et al., 2009](#)) is a free package for network visualization. It can be used as an alternative to cytoscape. It uses openGL to accelerate graphics and allows quick rendering and visualization of huge networks. Options to visualize

**FIG. 10.9**

Graphical User interface of Cytoscape app, NAViGaTOR and VisANT.

graphs in both 2D and 3D are available and we can combine all nodes into one “meta node”. NAViGaTOR also contains API for plugins and different data formats. Lasso selection method and bookmarking to support manual design and other network analysis operations (Fig. 10.9).

10.11.3 VisANT

VisANT (Hu et al., 2008) is a network visualization tool which can run in the browser or as a separate program. One of its interesting features is the name resolution feature, which tries mapping nodes in the network to different gene names, resulting in each protein coded by a gene to be depicted as a separate entity. This name-mapping feature is one of the easiest to use and understand as compared to many of the other software packages available. VisANT has been used on large datasets having a large number of nodes. Metagraphs can also be represented using this tool. In metagraphs, a single node can contain a subgraph. This tool is also integrated with online databases, containing a large number of interactions (Fig. 10.9).

10.11.4 CellDesigner

This is diagram editor tool used for drawing biochemical networks and gene regulatory networks. The user can search or alter networks in the form of process diagrams and stock the networks in standard for depicting these networks, known as systems biology markup language. This editor can connect the network with simulations. Users can see the network dynamics through a GUI (Funahashi et al., 2007). This editor is implemented in Java and integrated different packages.

10.11.5 Pathway Studio

Pathway Studio (Nikitin et al., 2003) is another free tool that uses NLP- based data extraction for gene-gene and protein–protein interactions. Pathway Studio contains databases of cellular pathways and protein interactions. This tool effectively scans biological words and terms occurring along with them. The database has a large library of interactions and proteins extracted from literature present on different organisms.

10.11.6 Gephi

A program that can work on large networks without the requirement of programming skills is Gephi (“Gephi - The Open Graph Viz Platform,” n.d.). Gephi can handle a large number of nodes and edges. However, this software requires a large amount of computational power. The advantages of using Gephi are that this software is free, can be used on different platforms and can use advanced algorithms as plugins. The limitation of this software is that it lacks the ability to analyze specific biological data. This tool can be used for visualizations, statistical analysis and enumerating.

10.12 Important properties to be inferred from networks

The first step of all network analysis processes is the construction of a network. Investigation of features in a network and their deviation from our expectations helps in better understanding the networks. Different computational and mathematical methods have been generated for the analysis of large networks for identification of the selected features.

10.12.1 Hubs

Node degree is an informative feature in network analysis. Interaction networks are usually present in a scale-free topology. In these networks, hub nodes are important for the functions and structure of these networks.

10.12.2 Bottlenecks

Nodes with high betweenness nodes are known as bottleneck nodes. This means, that these nodes are present in paths between many different nodes. Bottlenecks play an important role in the communication in a network, as they help in the flow of information among modules (Sub networks with dense connections). These nodes in a network are present on the major connections. Disturbances in the bottleneck nodes can lead the network into a disarray, due to presence of a low number of connections near bottlenecks. Bottleneck nodes are more important to networks as compared to the hub nodes, causing these bottleneck nodes to be targeted by pathogens.

10.12.3 Modules

Networks are modular and have communities in their structure. Specific modules are usually enriched to perform the usual biological functions. Therefore, detecting modules present in networks can help understand processes/functions that are not known in a pathway. Proteins related to a disease or disease having same phenotypes usually display interactions in modules of the disease. Therefore, we can detect modules that show enrichment in genes or proteins linked to a disease. Other module proteins which are not known to be linked to a disease, can be predicted to be associated to the disease. This method is now popularly being used to detect disease modules for a variety of diseases in humans. E.g., A network module controlling the heart development has been associated with heart disease (Fig. 10.10).

10.12.4 Bioinformatics tools to detect modules, bottlenecks and hubs

A broad range of tools are currently used for the fast detection of bottlenecks and hubs. An example is NetworkAnalyst, which can analyze networks based on gene expression. NetworkAnalyst can take a list of genes given by the user and interactions involved from InnateDB to find modules, betweenness centralities and degrees in the network. Cytoscape also contains applications that can help in network analyses. An example of such application is cytoHubba, which can detect bottlenecks and hubs in the network given as input to cytoscape. Cytoscape can be utilized along with the networks produced InnateDB. A wide range of tools have been created to detect network modules.

NetworkAnalyst has many analytical features that can detect network modules. NetworkAnalyst utilizes random walk algorithm to detect modules of high frequency nodes. This tool can build an edge weighted network, where node information like gene expression can be used to give weights. Cytoscape has many different applications for detection of modules. One such application is jActiveModules which detects connected parts of the network that have significant differences in

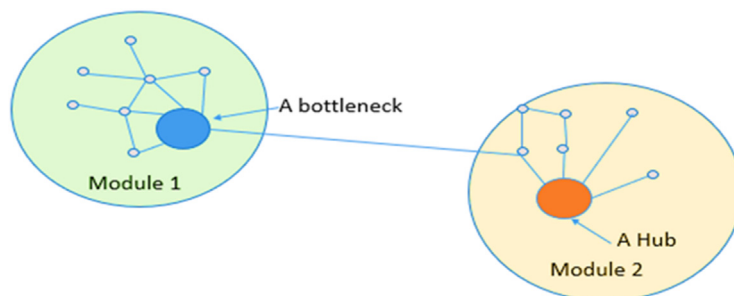


FIG. 10.10

Figure depicting hub, bottleneck and modules.

expression of genes. When modules associated with diseases need to be identified, other algorithms give a better performance, since proteins associated with diseases are not present in dense regions and nodes associated with diseases may be identified more accurately through connectivity significance. The DIAMOnD (Disease Module Detection) algorithm is a novel method to find disease modules on the basis of connectivity significance. Incompleteness of the interactome also restricts the disease modules which can be identified, and the small quantity of known disease linked proteins which can be used to detect the disease.

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